A

Report for

TD619: Energy Policy and Planning

On

Solar Mini-grid Design Using a Multidimensional Approach

Submitted by:

Team 5

Sanket Bafana | 21d170037
Sanket Mhaske | 21d170026
Aditya Vibhute | 21d170003
Pooja Mathary | 22b0358

Under the guidance of

Prof. Anand B. Rao

Centre for Technology Alternatives for Rural Areas (C-TARA)

Indian Institute of Technology, Bombay

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1. Introduction

Access to reliable electricity is crucial for the socio-economic development of rural communities. As part of our project on solar mini-grid design, we conducted a field visit to **Kahandolpohi**, a small village near Khardi, Mumbai, where **Pragati Pratishtan** has installed **a mini-grid system** as a **CSR initiative**. The mini-grid currently supplies electricity to **13 households**, providing an essential power source for daily activities.

The system's inverter has recorded the load demand and we have been given this data of three years. We analysed this data to understand consumption patterns, seasonal variations, and the operational performance of the existing system. We aimed to determine whether the current system capacity is sufficient to meet the village's energy demands.

To enhance the efficiency of the mini-grid, we replicated and modelled the collected data to optimise energy use and cost-effectiveness. Using HOMER software, we simulated different system configurations to find an optimal balance between energy generation, storage, and cost, ensuring a more sustainable and efficient solution for the village.

Additionally, we conducted interviews with local households to identify the challenges they face with the current system and incorporated their feedback into our improved model. This allowed us to address existing issues and propose a mini-grid design that is both technically and economically viable.

This report presents our analysis of the existing mini-grid, the challenges faced by the community, and our optimised design proposal using HOMER software that aims to provide a more sustainable, cost-effective, and reliable energy solution for the village.

2. Solar Mini-grid Techno-Economic Design

2.1. Solar Mini-Grid Technical Design

HOMER has a functionality where a series of load values with a step size of a minute can be used. The Excel file is converted to CSV and uploaded to HOMER. HOMER calculates the peak load, average load and plots the energy consumption pattern, and month-wise or seasonal variations.

Economic inputs like costs of components like panels, battery and converter, replacement cost and O&M for each, discount rate, and inflation rate are needed to proceed further. For simplicity, the replacement cost is assumed to be the same as the initial cost.

Generic solar panels, free size converter, and 2 different scenarios based on types of batteries - Lead acid and Lithium ion are considered. The life of a Li ion battery is 15 years, whereas a lead-acid battery is 10 years. Efficiency is also higher than lead-acid battery, however, the cost of Li ion is double that of lead-acid. So, to find out which scenario optimises the system (gives the least value of Levelised Cost of Electricity), the simulation is performed.

2.1.1 Technical Specifications of the System (Collected during field visit at Kandolpohi):

Table 1: Technical specifications of system installed at Kandolpohi

Components	Rating	Number
Panel	320 W	12
Battery	12 V, 200 Ah	4
Inverter	3500 VA	1

2.1.2 Data Analysis

The folder has data in the form of CSV files for each day for 3 years, though not completely for 2018 and 2020. The year 2019 has complete data for each day of each month. The months of 2018 and 2020, where data is absent, are ignored while calculating the average load curve for a year.

The data of an individual year is combined using Microsoft Excel into a single Excel file and the Solar Power Total (kW), which represents the load demand, is filtered out. The three different load curves of three years are then averaged out, and a final load series with a time step of a minute for an entire year is made as shown below.

Table 2: Load series with a time step of a minute for an entire year

2018	2019	2020	Average of three years
0.25	0.46	0.14	0.283333
0.28	0.44	0.15	0.29
0.28	0.4	0.15	0.276667
0.27	0.38	0.15	0.266667
0.27	0.43	0.16	0.286667
0.27	0.53	0.16	0.32
0.27	0.53	0.18	0.326667
0.27	0.49	0.18	0.313333
0.23	0.46	0.18	0.29
0.2	0.43	0.18	0.27
0.2	0.38	0.14	0.24
0.2	0.33	0.11	0.213333
0.2	0.29	0.13	0.206667
0.2	0.3	0.13	0.21
0.19	0.33	0.15	0.223333
0.19	0.33	0.17	0.23
0.19	0.33	0.2	0.24
0.19	0.34	0.2	0.243333
0.18	0.34	0.17	0.23
0.19	0.29	0.2	0.226667
0.19	0.25	0.23	0.223333
0.2	0.27	0.18	0.216667
0.2	0.27	0.15	0.206667
0.2	0.25	0.2	0.216667
0.21	0.23	0.24	0.226667

The peak load is found to be **2.42 kW** occurring in the month of July for some minutes using this minute-wise load curve, which the inverter has to handle.

From this average load curve, the hourly load curve for a typical day in each month is obtained as shown below (all values in kW).

Table 3: Hourly load for a typical day in each month

Yearly Load Data

Weekdays Weekends

Hour	January	February	March	April	May	June	July	August	September	October	November	December
0	.36	.24	.13	.12	.11	.04	.01					
1	.3	.2	.12	.12	.1	.01						
2	.22	.17	.09	.09	.06							
3	.18	.15	.08	.08	.04							
4	.15	.1	.04	.04	.07	.05	.03	.06	.12	.14	.09	.05
5	.13	.07		.04	.18	.17	.13	.26	.5	.66	.63	.53
6	.11	.05	.01	.21	.36	.36	.33	.47	.68	.77	.84	.85
7	.17	.08	.12	.35	.48	.46	.58	.56	.63	.61	.63	.7
8	.46	.37	.31	.49	.57	.51	.8	.6	.58	.47	.47	.5
9	.41	.39	.36	.46	.45	.49	.87	.63	.47	.36	.35	.38
10	.28	.28	.37	.37	.35	.4	.84	.63	.37	.3	.31	.32
11	.22	.21	.28	.29	.29	.34	.82	.6	.32	.27	.26	.27
12	.18	.16	.23	.24	.24	.31	.7	.55	.28	.23	.22	.24
13	.15	.15	.19	.21	.22	.37	.59	.48	.23	.19	.2	.2
14	.14	.13	.17	.18	.23	.45	.48	.38	.21	.18	.19	.17
15	.13	.12	.15	.15	.25	.56	.37	.25	.12	.07	.03	.02
16	.12	.09	.13	.13	.29	.55	.25	.1				
17	.08	.1	.17	.18	.36	.5	.14	.03				
18		.12	.31	.33	.37	.42	.09					
19		.18	.37	.43	.37	.35	.06					
20	.02	.24	.28	.36	.26	.25	.06					
21	.06	.39	.2	.26	.21	.15	.04					
22	.13	.36	.16	.19	.16	.12	.03					
23	.31	.26	.14	.15	.13	.07	.02					

A typical daily electricity consumption (month wise) is calculated by using the hourly load curve for a typical day in each month.

Table 4: A typical daily electricity consumption (month wise)

Months	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Total (kWh)	4.31	4.5	4.22	5.45	6.11	6.88	**7.58**	5.7	4.63	4.39	4.31	4.28

The peak daily electricity consumption occurs in July, which is **7.58 kWh,** which the system has to cater to.

2.1.3 Design Calculations

The peak load is 2.42 kW, and the maximum consumption is 7.58 kWh. So we will consider that the system must cater to this.

Assumptions:

Battery efficiency = 0.75

Inverter efficiency = 0.9

Charge controller efficiency = 0.95

Depth of Discharge of battery (DoD) = 0.8

Other data used:

Same specifications as those of the installed system at Kahandolpohi are used.

Panel	Power	320 W
	Impp	8.63 A
	Vmpp	37.1 V
Battery	Voltage	12 V
	Capacity	200 Ah

Sizing of components:

1. **Inverter Sizing:**

   Peak load = 2.42 kW

   Power factor = 0.7

   Rating of Inverter = Peak load/ Power factor = 2.42 / 0.7 = **3.45 kVA**

2. **Battery Sizing :**

   Parameters:

   DoD = 0.8, Rated capacity = 200Ah, Voltage = 12 V , Autonomy = 1 day, Efficiency = 0.75

   <u>1. Energy Input to Inverter</u>

   Energy Input to Inverter = Units per day ÷ Inverter Efficiency

   = 7.58 ÷ 0.9 = 8.4222 kWh/day

   <u>2. Total Ah Required</u>

   Total Ah Required = (Energy Input to Inverter × 1000 × (1 + Autonomy)) ÷ (System Voltage × DoD × Battery Efficiency)

   = (8.4222 × 1000 × 2) ÷ (48 × 0.8 × 0.75) = 585 Ah

<u>3. Number of Batteries Required (Total)</u>

Number of Batteries = Total Ah Required ÷ Battery Rating

= 585 ÷ 200 = 2.9 batteries = approx. 3 batteries

<u>4. Batteries in Series</u>

Batteries in Series = System Voltage ÷ Battery Voltage

= 48 ÷ 12 = 4 batteries per string

<u>5. Number of Strings (Parallel)</u>

Number of Strings = Total Number of Batteries ÷ Batteries in Series

= 3 ÷ 4 = 0.75 = approx. 1 string

Therefore, three batteries will be sufficient with 1 day of autonomy. But to match the system voltage of 48 V, **four batteries each of 12 V are to be placed in series**.

3. **Solar Panels Sizing:**

Parameters:

Panel Power (P) = 320 W

Impp (Current at Maximum Power Point) = 8.63 A

Vmpp (Voltage at Maximum Power Point) = 37.1 V,

Battery efficiency = 0.75

Charge controller efficiency = 0.95

System Voltage = 48 V

Sunshine hours = 5.5 hrs

Case 1) Without Autonomy

1. Total Current Needed from PV Panels

I = Load Power (W) / System Voltage (V) = 1920 / 48 = 44.78 A

2. Number of Panels in Parallel

Panels in parallel = I / Impp = 44.78 / 8.63 ≈ 5.19 ⇒ 6 panels (rounded up)

3. Number of Panels in Series

Panels to be placed in series = System Voltage / Vmpp = 48 / 37.1 ≈ 1.29 ⇒ 1 panel (rounded down as system voltage is 48 V)

4. PV System Capacity

Total Capacity = Total Number of Panels × Panel Power = 6 × 320 = 1920 W = **1.92 kW**

Case 2) With 1 Day Autonomy

1. Total Current Needed from PV Panels

I = Load Power (W) / System Voltage (V) = 3520 / 48 = 89.55 A

2. Number of Panels in Parallel

Panels in parallel = 89.55 / 8.63 ≈ 10.38 ⇒ 11 panels (rounded up)

3. Number of Panels in Series

Panels to be placed in series = 48 / 37.1 ≈ 1.29 ⇒ 1 panel

4. PV System Capacity

Total Capacity = 11 × 320 = 3520 W = **3.52 kW**

Case 3) With 2-Day Autonomy

Similarly to above, the PV system capacity with 2 days autonomy turns out to be:

Total Capacity = **5.12 kW** (16 panels connected in parallel to each other)

2.1.4 Comments on Comparison with Existing System

The system we designed (3.52 kW), considering autonomy of 1 day, turned out to be very similar to that installed at Kahandolpohi (3.84 kW). The only difference is in the number of solar panels (one less than actually installed) and the rating of an inverter (difference of 50 VA only), as the data will slightly change the design.

2.2 Mini-grid Economics

2.2.1 Details of various costs

Homer was used to optimise to minimise NPV. The discount rate is considered to be 10% [1][2], and the Inflation rate is taken to be the average of the five years' data that was provided, hence 5.53% [3].

Here we are assuming that the system fixed cost consists of the meters, mounting structure and distribution system cost (which includes electricity cables mainly). Assuming linear variation passing through the origin, the value of the system fixed cost was interpolated to be 86,000 Rs [4], and the O&M cost for the same was assumed to be 2% [4] of the capital cost, which comes out to be 1,720 Rs.

Table 5: Inputs used for optimisation

Components considered for optimisation	Capacity	Life	Capital cost	O&M cost
WAAREE All Black 405Wp 132 Cells 24 Volts Mono PERC Solar Panel [5] (Monocrystalline solar panel)	0.405 kWp	25 years	8000 Rs	320 Rs/year
LUM 12170P-Mono Facial [6] (Polycrystalline solar panel)	0.170 kWp	25 years	6500 Rs	280 Rs/year
CAML 48 V 100 Ah, 5 kWh Lithium Battery - Rack Mount [7] (Li-ion battery)*	5 kWh	15 years	1,08,000 Rs	0 Rs/year
Solar Battery 200 Ah - LPTT12200H [8] (Lead-acid battery)	2.4 kWh	10 years	21,500 Rs	0 Rs/year
Microtek HI-END MPPT PCU 5KVA/48V [9] (Inverter)*	5 kVA	15 years	80,000 Rs	0 Rs/year

* This value was used to calculate Rs./kWh or Rs./kW value of Li-ion battery or inverter, since a 5 kWh or 5 kVA model was not available so we used a 1 kWh generic lithium ion model and generic free converter for optimisation, therefore the quantity in the table mentioned below is based on 1 kWh battery and the generic free converter

Table 6: Scenarios considered along with total capacity

Scenario no.	PV panel	Capacity	Battery	Quantity	Converter
1	Monocrystalline solar panel	10.6 kW	Lead-acid battery	5	Inverter
2	Polycrystalline solar panel	4 kW	Lead-acid battery	20	Inverter

3	Monocrystalline solar panel	11.4 kW	Li-ion battery	10	Inverter
4	Polycrystalline solar panel	6 kW	Li-ion battery	30	Inverter

2.2.2 Economic Calculations

Please refer to the spreadsheet shown below for the economic calculations done :-

https://docs.google.com/spreadsheets/d/1xVv0c_J_p0M9rgHTYRngD5KWzoa3NxI3PvI7kxtQTtc/edit?usp=sharing

Scenario no.	NPV (Rs)	LCOE (Rs/kWh)	CAPEX (Rs)	Operating cost (Rs/yr)
1	-6,27,326.30	2.38	3,42,162.71	10,095.30
2	-12,06,306.14	12.64	6,04,221.17	7,837.64
3	-7,38,152.67	2.30	4,51,294.81	11,912.59
4	-14,23,677.92	9.94	9,96,745.1	29609.23
Actual scenario				

2.2.3 Homer Results

Table 7: Results from HOMER optimisation

Scenario no.	NPV (Rs)	LCOE (Rs/kWh)	CAPEX (Rs)	Operating cost (Rs/yr)
1	6,92,642.5	24.08	4,28,915.8	17304.94
2	12,13,984	42.21	6,90,274.5	34364.21

3	8,07,031.6	28.06	5,41,375.8	17431.54
4	14,47,989	50.34	9,96,745.1	29609.23

Inferences:

1. With the lowest Levelized Cost of Energy (LCOE) (₹24.08/kWh) and Net Present Value (NPV) (₹6,92,642.5), Scenario 1 (monocrystalline solar panel + lead-acid battery) is ultimately the most economical choice. With the highest NPV (₹14,47,989) and LCOE (₹50.34/kWh), scenario 4 (polycrystalline solar panel + Li-ion battery) is the least desirable from an economic standpoint.

2. Monocrystalline panels outperform polycrystalline panels in cost-effectiveness despite their higher upfront cost. Li-ion batteries are more expensive initially but may provide long-term benefits, though this is not reflected in the current economic assessment. This is possible because there was no definite model available for Li ion battery simulating its life span, efficiency, charge/discharge rate, etc.

3. If the objective is cost minimisation, Scenario 1 (Monocrystalline + Lead-acid) is the most economical. If long-term battery lifespan is prioritised, Scenario 3 (Monocrystalline + Li-ion) could be a viable alternative despite a higher upfront cost.

2.3. Comparison of Design with Existing System:

Table 8: Comparison of Design with Existing System

COMPONENTS		EXISTING SYSTEM	DESIGNED SYSTEM
PANELS	RATING	320 W	320 W
	NUMBER	12	11
BATTERY	RATING	12 V, 200 Ah	12 V, 200 Ah
	NUMBER	4	4
INVERTER	RATING	3500 VA	3450 VA
	NUMBER	1	1

The system we designed (3.52 kW), considering autonomy of 1 day, turned out to be very similar to that installed at Kahandolpohi (3.84 kW). The only difference is in the number of solar panels (one less than actually installed) and rating of the inverter (difference of 50 VA only), as the noise in the data will slightly change the design.

3. Social Dimension Consideration for Mini-grid Design

The introduction of new technology or the transition from existing systems to innovative solutions within a society presents significant challenges. Despite offering similar technological features and pricing, certain technologies succeed in some regions while failing in others. This disparity often arises because technologists and stakeholders involved in implementation overlook the crucial role of social factors. Sustainable technology adoption is not merely about introducing new equipment; it requires an understanding of how communities with diverse socio-cultural backgrounds interact with technology. How individuals and communities engage with technological interventions influences their long-term viability and effectiveness in everyday life. Therefore, when introducing a solar mini-grid system, it is essential to consider social factors during the design phase to minimise the risk of failure and ensure inclusivity.

3.1 Social factors to consider during the design phase

1. **Community Demographics and Diversity**

- Mini-grids operate centrally; community composition directly impacts system design.
- Treating all users as homogeneous overlooks key socio-cultural and economic differences.

  In Homogeneous Societies:
  - Similar income levels, culture, and energy needs.
  - Uniform consumption simplifies demand forecasting and tariff design.
  - Easier to plan for capacity and future expansion.

  In Heterogeneous Societies:
  - Wide disparities in income, expectations, and ability to pay.
  - Higher-income groups consume more energy and are willing to pay more.
  - Few social groups have the perception that they are entitled to get free electricity from the government.[10]
  - Tariff system design becomes challenging and may lead to inequity.

Proposed Solution:

- Analyse consumption patterns across social groups.
- Offer free electricity up to a threshold (e.g., average of the lowest 10% users).
- Charge for usage above this threshold to ensure equity and affordability.
- This would encourage the use of mini-grids rather than completely shifting to the main grid electricity.

2. Social Acceptance

- In Kahandolpohi, residents accepted mini-grids due to a lack of alternatives.
- Primary concern: system reliability.

Contrasting Example (Bihar):

- Solar is perceived as "fake electricity"; the community preferred a government grid supply.[10]
- Misunderstandings about solar capacity and sustainability.
- To create awareness about the technology, targeted community engagement programs are essential. These could include demonstration projects, user training, and regular interaction with local leaders to build trust.

Technical Solution:

- Use modular, scalable technology for demand growth (e.g., stackable batteries [12], modular inverters [11]).
- Plan for flexible expansion without large upfront oversizing.

3. Post-Installation Maintenance and Service

- Long-term functionality depends on reliable operation and maintenance (O&M).
- In Kahandolpohi:
  - Battery failure left the community without electricity.
  - No response from the NGO that installed the system.
  - The community had to self-fund battery replacement after 6 months of blackout.

Key Issues:

- NGOS often provide a one-time installation without maintenance plans.
- No structured O&M leads to system failure and financial waste.

Recommended Approach:

- Establish community-based O&M models:
  - Form local energy committees for daily operation and coordination.

    - Train local technicians to handle minor repairs.
    - Create community funds with small monthly household contributions.
  - Enhances resilience, local ownership, and system longevity.

4. **Ensuring Consistency in Tariff Collection**
   - Reliable tariff collection ensures funds for maintenance and system sustainability. This was successfully achieved in Kahandolpohi as the person was from the same community.
   - Barriers:
     - Reluctance to pay unless benefits are perceived.
     - Non-payment despite ability leads to unfairness.
     - Lack of trust in fund management discourages contribution.

   <u>Best Practices:</u>
   - Use trusted community members for tariff collection to ensure transparency.
   - Promote community ownership and accountability.
   - Establish clear payment policies and regular financial updates to build trust.
   - Encourage fairness and collective responsibility.

3.2 Suggested Parameters for Measurement & Ways to Measure During Mini-Grid Operation

1. Community Demographics

<u>Parameter: Homogeneity vs. Heterogeneity</u>

How to Measure:

- Conduct household surveys to gather data on income levels, occupations, and cultural diversity.
- Analyse electricity usage patterns among different socio-economic groups.
- Organise community meetings to assess participation and concerns about energy entitlements.

2. Social Acceptance

<u>Parameter: Resistance Due to Misconceptions</u>

How to Measure:

- Conduct awareness surveys before and after education campaigns to measure knowledge about solar energy.
- Track usage trends to see how many households consistently use power vs. those who hesitate. Organise community feedback sessions to address concerns about "fake electricity."

<u>Parameter: Scalability & Expansion Adoption</u>

How to Measure:

- Monitor system expansion requests from households seeking increased energy capacity.
- Analyse load growth trends to understand demand changes as modular components are added.
- Record upgrade investments by tracking purchases of additional appliances or energy capacity increases.

3. Post-Installation Maintenance Service

Parameter: System Downtime Due to Maintenance Issues

How to Measure:

- Maintain service logbooks to track breakdown frequency and causes.
- Measure repair response time for addressing reported issues.
- Conduct user satisfaction surveys to assess maintenance reliability.

Parameter: Effectiveness of Community-Based O&M Model

How to Measure:

- Keep technician training records to count trained local maintenance personnel.
- Monitor community fund contributions for regularity and amount collected for repairs.
- Track issue resolution rates to determine the percentage of reported problems successfully resolved.

4. Ensuring Consistency in Tariff Collection

Parameter: Tariff Payment Compliance

How to Measure:

- Maintain payment records to track on-time vs. late payments.
- Identify defaulter rates by analysing recurring missed payments.

Parameter: Trust in Tariff Collection System

How to Measure:

- Organise transparency meetings to provide community updates on fund usage.
- Conduct user feedback surveys to gauge trust levels in the tariff collection committee.
- Maintain conflict records to document disputes related to unfair billing or tariff setting.

4. Institutional Dimension Consideration for Mini-grid Design

Mini-grids are an effective solution for providing reliable electricity to rural villages in India. The success of a solar mini-grid system depends on its technical design and the institutional framework governing its operations. Let's see Standard Operating Procedures (SOPs) for sustainable mini-grid operations, focusing on business models, governance, measurement parameters, and operational best practices.

Institutional Factors Driving Mini-Grid Operations

1. **Regulatory Framework:** Policies and subsidies that influence mini-grid deployment
2. **Financial Viability:** Availability of funding, tariffs, and return on investment
3. **Community Engagement:** Involvement of local populations in governance & decision-making
4. **Technical Expertise:** Presence of skilled personnel for installation, maintenance & operation
5. **Infrastructure Development:** Availability of roads, logistics, and supply chains

4.1 Different Business Models based on Governance:

Business Model	Structure	Advantages
Community-Base Model	Owned and operated by a community cooperative	High local involvement Increased sustainability
Private Sector Model	Fully owned and operated by a private entity	Efficient management Investment potential
Public-Private Partnership (PPP) Model	Joint venture between government and private players	Shared risk and investment Government backing
Anchor - Commercial-Community (ABC) Model	Prioritises electricity supply to an anchor load (e.g., telecom towers), ensuring financial stability	Stable revenue from anchor clients Potential for rural expansion
Hybrid Model (Recommended)	Combination of community, private, & government stakeholders	Balances local engagement, and financial sustainability

4.2 Business Models: Overview & Challenges:

1. **Community-Based Model**

   A cooperative-run mini-grid where residents manage and maintain the system

   - ◆ **Example:** A community-run solar mini-grid in Katamanda village, Chipangali district, supplies electricity to households, schools, and a rural health center
   - ◆ **Challenges:**
     - Difficulty in maintaining revenue collection and operational costs
     - Technical Expertise – Lack of skilled personnel for maintenance
     - Slow decision-making and lack of accountability

2. **Private Sector Model**

   A for-profit model where a private company invests in, builds, and operates mini-grid.

   - ◆ **Example:** TPRMG, a subsidiary of Tata Power, aims to roll out 10,000 microgrids, with 200 commissioned in Uttar Pradesh and Bihar, and a pilot in Odisha
   - ◆ **Challenges:**
     - Affordability – High tariffs make power expensive for rural consumers
     - Prioritises industrial clients over household electrification
     - Risk of customer loss if the national grid arrives with lower tariffs

3. **Public-Private Partnership (PPP) Model**

   A collaboration between the government & private companies to develop & operate grids

   - ◆ **Example:** Kenya government and private companies are collaborating to develop and operate off-grid electricity systems, particularly through initiatives like the Kenya Off-grid Solar Access Project (KOSAP)
   - ◆ **Challenges:**
     - Government approvals slow down implementation
     - Conflicting Objectives – Private firms aim for profit, while the government focuses on social impact

4. **Anchor-Commercial-Community (ABC) Model**

   Focuses on supplying electricity first to anchor clients (e.g., telecom towers) to generate stable revenue, then extends power to businesses and homes.

   - ◆ **Example:** Project by OMC Power, located in Bilgram in the state of Uttar Pradesh.

- ◆ **Challenges:**
    - Anchor loads (e.g., telecom towers) are prioritised over household electrification
    - High Initial Investment – Requires substantial capital for infrastructure and securing anchor clients

5. **Hybrid Model (Recommended)**
   A combination of ABC and PPP models, integrating stable revenues from anchor clients, government incentives, and community engagement
   - ◆ **Challenges:**
     - Complex Governance – Coordination between the community, the private sector, and the government can be difficult

4.3 Parameters for Measurement in Mini-Grid Operations:

Key performance parameters must be measured regularly to ensure the sustainable operation of mini-grids.

a. **Technical Parameters:** These indicators ensure the reliability and efficiency of mini-grid
   - Availability of Power (%): Uptime of the mini-grid (hours of availability per day)
   - Load Factor: Ratio of actual energy consumption to maximum possible energy output
   - Voltage & Frequency Stability: Ensuring power quality remains within acceptable limits
   - Battery Storage Performance: Charge/discharge cycles, efficiency, and lifespan
   - Smart Meters & IoT Sensors: For real-time tracking of energy consumption

b. **Financial Parameter:** These indicators track economic sustainability.
   - Tariff Collection Efficiency (%): Percentage of billed amount successfully collected
   - Return on Investment (ROI): Financial viability based on capital & operational costs
   - Cost per kWh (₹/kWh): Total operational & maintenance costs / units of energy sold
   - Revenue Diversification Index: Measures revenue stability across different customer segments (households, businesses, anchor clients)

c. **Social Parameters:** These indicators measure the impact on local communities.

- Household Electrification Rate (%) – Percentage of households in the village connected to the mini-grid
- Customer Satisfaction Index – Measured via periodic surveys on power quality & reliability
- Income Growth of Beneficiaries (%) – Economic improvement of local businesses due to a reliable power supply
- Surveys & Interviews – To assess social and financial impact

The **Hybrid Model** is often the most effective and sustainable for mini-grid operations, particularly in rural India. This model combines elements of Public-Private Partnership (PPP) and Community-Based models, ensuring financial viability, community ownership, and long-term sustainability.

Reasons:

- **Diversified Funding** – Combines government subsidies, private investments, and community contributions, reducing financial risks
- **Efficient Management** – The Private sector brings technological expertise, while local community involvement ensures better acceptance and maintenance
- **Affordable & Reliable Power** – Tariffs are balanced between cost recovery and affordability through government support
- **Community Empowerment** – Involves local stakeholders in governance, ensuring better accountability and service quality

5. Environmental Dimension Consideration for Mini-grid Design

5.1 Discussion on Existing Literature

Solar systems have the potential to reduce carbon emissions and provide sustainable and green energy for the rapidly growing demand. Solar mini-grids can displace more polluting energy sources like diesel generators, leading to a reduction in carbon dioxide emissions. By replacing fossil fuel-based power generation, solar mini-grids can significantly reduce air pollution, particularly in areas reliant on diesel generators. By reducing greenhouse gas emissions and promoting renewable energy, solar mini-grids play a role in mitigating climate change. To understand the impact of the solar mini grid on the environment, we need to look at how the CO2 emissions are much lower when compared to the conventional thermal power plant, if the same electricity is produced from them. It provides access to clean and renewable energy in remote or underserved areas, improving the quality of life and reducing reliance on harmful energy sources like firewood.[17]

Like every coin has two sides, solar technologies also have some negative impacts on the environment. The production phase of solar cells involves toxic chemical substances, with solar cell batteries containing heavy metals like lead, cadmium, nickel, cobalt, and manganese. These materials pose threats to natural resources due to their short lifespan and potential for environmental contamination. While solar cells do not emit substances during operation, emissions can occur during manufacturing and transport. Additionally, large-scale use of solar energy can lead to water pollution due to the use of anti-icing agents, anti-corrosion agents, and herbicides, which may enter water systems during maintenance activities. While solar energy offers significant environmental benefits over conventional energy sources, it's crucial to address and mitigate its potential negative impacts through careful planning, sustainable practices, and technological advancements.[18]

5.2. Environmental Impact Assessment of Mini-grids

Life Cycle Approach (Cradle to Grave) analysis is used to determine the impact of a solar mini grid on the environment by calculating the carbon footprint of each component. In the end, how many emissions can be avoided in one year with the help of a mini grid compared to if the same electricity is generated using a thermal power plant.

5.2.1 Carbon footprint calculation:

For simplification of calculation, we will use a 1 kW system and determine its carbon footprint. [19]

Capacity: 1 kW

a) Solar Panels:

   Carbon emission factor = 0.059 : approx. 0.06 Kg per kWh

   Units generated per year = 1500

   Emissions per year = 1500×0.06 = **90 Kg**

   Emissions lifetime = 90×25 = **2250 Kg**

b) Battery: [20]

Battery Capacity = 1.00 kWh

Emission Factor = 89.00 kg CO_2 per kWh

No. of Batteries = 8 (Considering replacement one time)

Emissions lifetime = 89 × 8 = **712 Kg**

Emissions per year = 712 / 25 = **28 Kg**

c) Inverter: [21]

A paper has mentioned the carbon footprint of 3.6 kW inverter, which has been used here.

Emissions lifetime = 1198 Kg

Emissions per year = 1198 / 25 = **48 Kg**

d) Mounting structure and cables:

The emissions due to this are negligible compared to the other components of the mini grid and hence these can be neglected but to understand how small they are a rough calculation is being done here.

Consider :

Steel = 80 Kg

Concrete = 100 Kg

Therefore,

Steel emissions = 80 × 3 = 240 Kg

Concrete emissions = 100 × 0.3 Kg = 30 Kg

Total CO2 emissions = 270 Kg

Per year CO2 emissions = 270 / 25 = **10.8 Kg**

5.2.2 CO2 avoided by mini grid:

Total emissions (solar mini grid) per year = Solar panel + Battery + Inverter + Mounting structure = 90 + 28 + 48 + 10.8 = **176.8 Kg**

Total emissions (solar mini grid) = 176.8 × 25 = 4420 Kg

<u>If same electricity is produced by coal based thermal power plant:</u>

Emission factor: 0.82 Kg CO_2 per kWh

Generation in one year by solar mini grid = 1500 kWh

CO_2 emission (thermal power plant) = 1500 × 0.82 = 1230 Kg/year

CO_2 avoided per year = (1230 - 176.8) = 1053.2 Kg

CO_2 avoided in Lifetime = 1053.2 × 25 = 26.33 tons

No. of years required by minigrid to compensate for CO2 emissions = 4420/ 1053 = **4.2 years**

Solar mini grid will compensate for its CO2 footprint in just 4.2 years and can operate with zero GHG emissions for the rest of its life.

6. Recommendations for Operational Sustainability of Mini-grids

When the main grid reaches areas previously served by mini-grids, the sustainability of these systems can be threatened due to factors like customer migration to the (often subsidised) main grid, regulatory uncertainties, and reduced revenue.

Here are some ways to save the mini grid and keep it operating

- Technical Integration with the Main Grid and sell power to the Main Grid
- Anchor clients: Secure long-term power purchase agreements with institutions, telecom towers, or industries
- Productive use of energy: Promote the establishment of cold storage units, agro-processing plants, water pumping systems, etc.
- Develop EV charging stations for two- and three-wheelers to capitalise on growing e-mobility trends
- Battery Storage Integration: Add battery energy storage systems (BESS) to manage peak demand and provide load balancing

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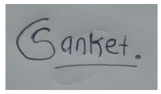
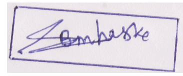
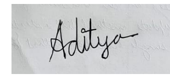
Appendices

Declaration of individual contribution in Mini-Grid Project

Group No: 05 Date: 18/04/2025

No	Criteria	Sanket Bafana 21d170037	Sanket Mhaske 21d170026	Aditya Vibhute 21d170003	Pooja Mathary 22b0358	Total
1	Communication & cooperation	25%	25%	25%	25%	**100%**
2	Quality of individual work	25%	25%	25%	25%	**100%**
3	Contribution to brainstorming & discussion	25%	25%	25%	25%	**100%**
4	Presentation and Report preparation	25%	25%	25%	25%	**100%**

5	Presentation delivery and Q&A	25%	25%	25%	25%	**100%**
	I hereby declare that I have reviewed and accept the individual contributions documented for all group members.					
Signature						

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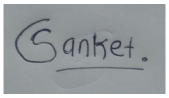

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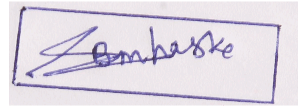

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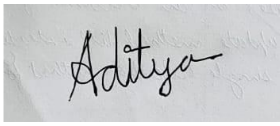

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Pooja Mathary | 22b0358 | 18/04/2025